

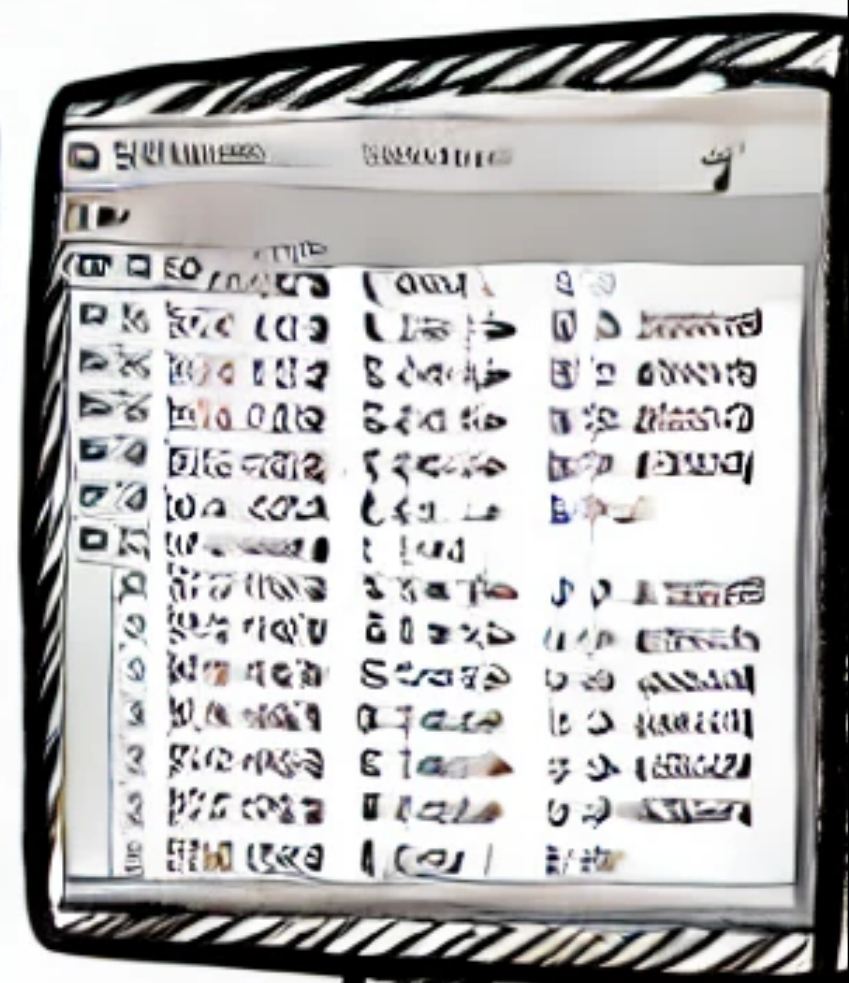
Introduction to OpenTelemetry

Josh Lee, Open Source Developer
Advocate @ Altinity

Modern Application Debugging

How do we debug our applications?

1. Mainlining from syslog intravenously
2. Deciphering hieroglyphs on dashboards
3. Asking Google (or ChatGPT)



A Typical Request Log

```
2024-07-01 09:35:34 GET /home 200 ...
```

Adding Duration

```
2024-07-01 09:35:34 231ms GET /home 200
```

Tada! Service-Level Metrics:

- Requests (Throughput)
- Errors
- Duration (Latency)

```
2024-07-01 09:35:34 231ms GET /home 200
```

Metrics are the foundation of

- alerting
- SLI/SLOs
- resource monitoring

We have monitoring and logging...

Back to our logs:

```
2024-07-01 09:35:34 231ms GET /home 200
```

Back to our logs:

```
Request:123 2024-07-01 09:35:34 231ms  
GET /home 200
```

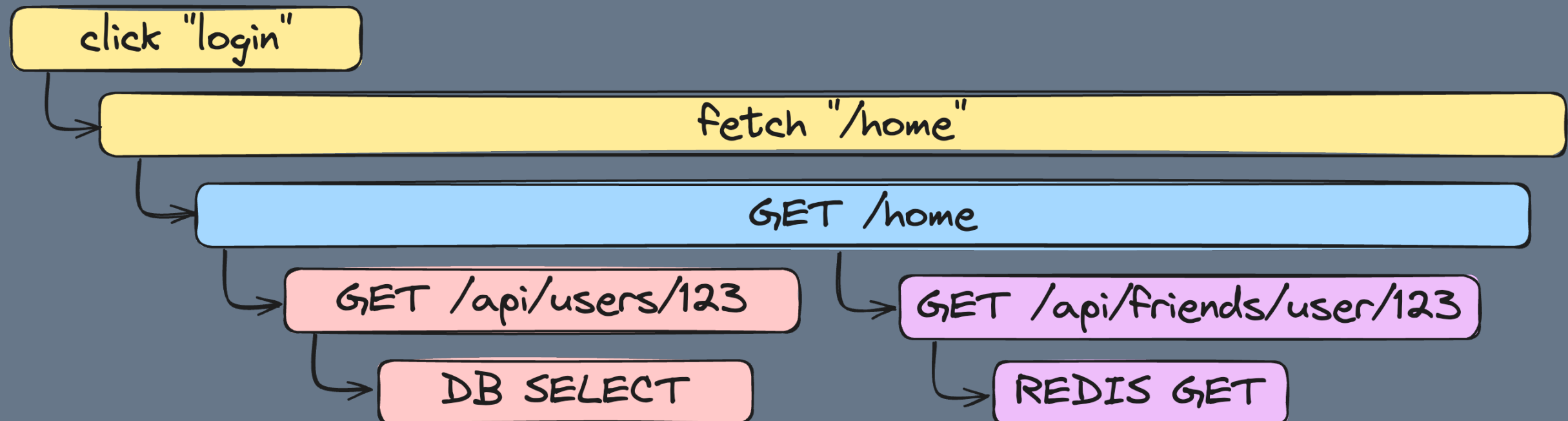
Connecting the trace:

```
Trace:123 Span:123 2024-07-01 09:35:34  
231ms GET /home 200
```

```
Trace:123 Span:456 ParentSpan:123 2024-  
07-01 09:35:34 201ms GET /api/users 201
```

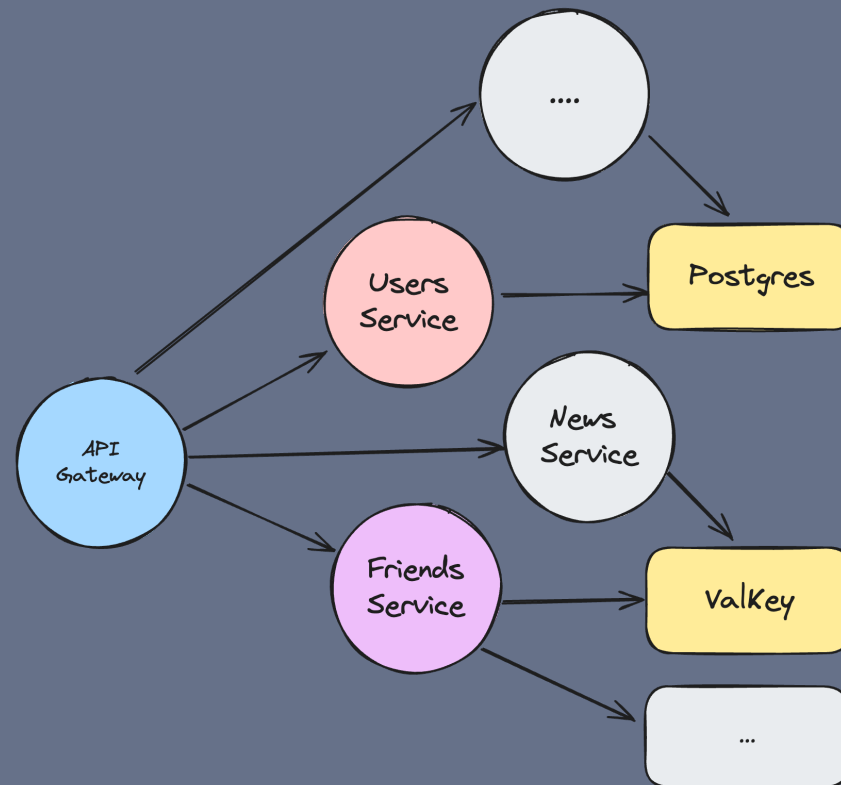
Traces

Preserve the complete history of each request



Traces

Encode the topology of the application in their structure



Metrics: Is there a problem?

Logs: What is the problem?

Traces: *Where* is the problem?

Traces

Can have errors and other signals attached to them for complete context

Traces

Can be analyzed in
aggregate to derive metrics

Traces

Can contain custom attributes (like `productId` or `orderValue`)

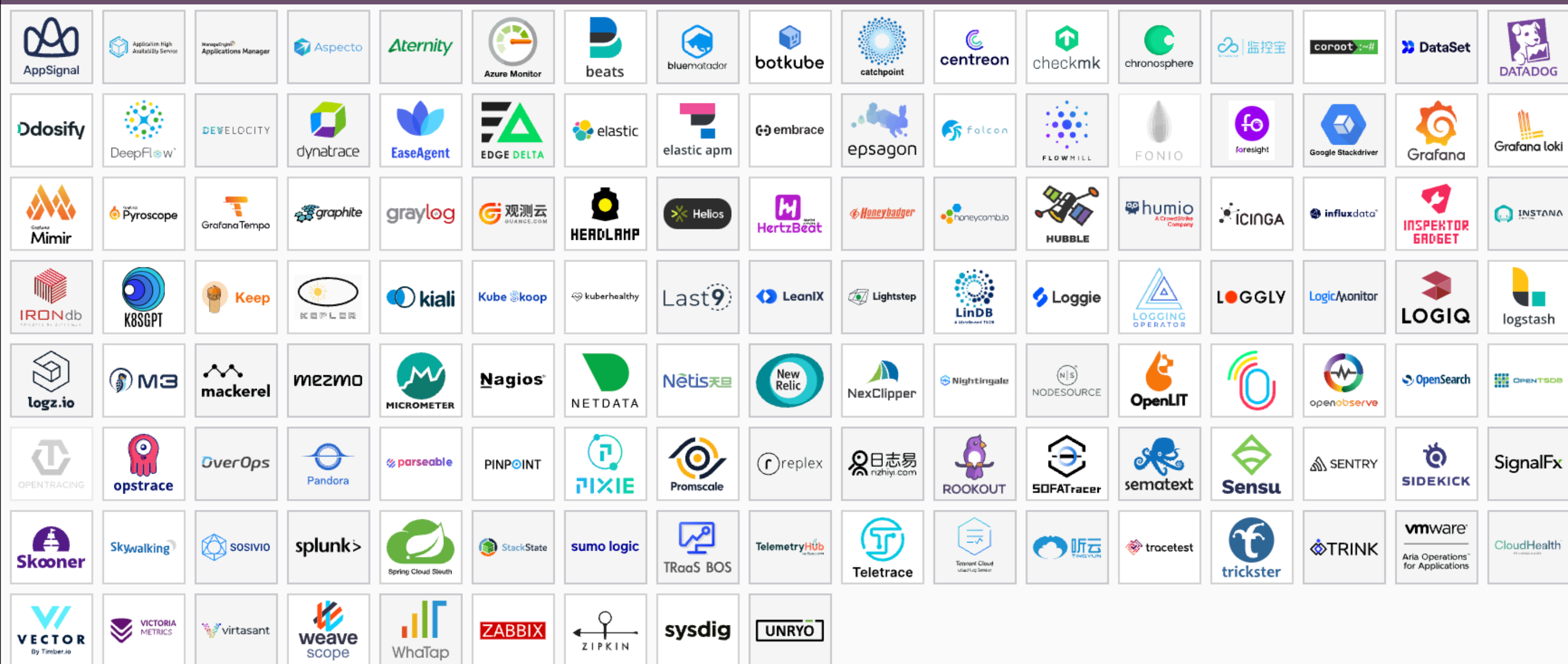
Traces

**Provide a user-centric way
to understand your system.**

Observability

Connects end users experiences to the teams that are responsible for them.

**Isn't this about
OpenTelemetry?**



Why OpenTelemetry?

**We need a
unifying
standard**

What is OpenTelemetry?

- community
- standards
- tools

OpenTelemetry Community

- CNCF GitHub/Slack
- Special Interest Groups
- Events & Meetups 🖐️



OpenTelemetry Standards

- W3C Trace Context Specification
- OTLP Specification
- OpenTelemetry API
- Semantic Conventions

OpenTelemetry Tools

- Language-Specific APIs & SDKs
- The OpenTelemetry Collector
- Kubernetes Operator
- BYOB

In your code*

OpenTelemetry
APIs & SDKs

OpenTelemetry SDKs

Language	Traces	Metrics	Logs
<u>C++</u>	🟢	🟢	🟢
<u>C#/.NET</u>	🟢	🟢	🟢
<u>Erlang/Elixir</u>	🟢	🚧	🚧
<u>Go</u>	🟢	🟢	🟡
<u>Java</u>	🟢	🟢	🟢
<u>JavaScript</u>	🟢	🟢	🚧
<u>PHP</u>	🟢	🟢	🟢
<u>Python</u>	🟢	🟢	🚧
<u>Ruby</u>	🟢	🚧	🚧
<u>Rust</u>	🟡	🚧	🚧
<u>Swift</u>	🟢	🚧	🚧

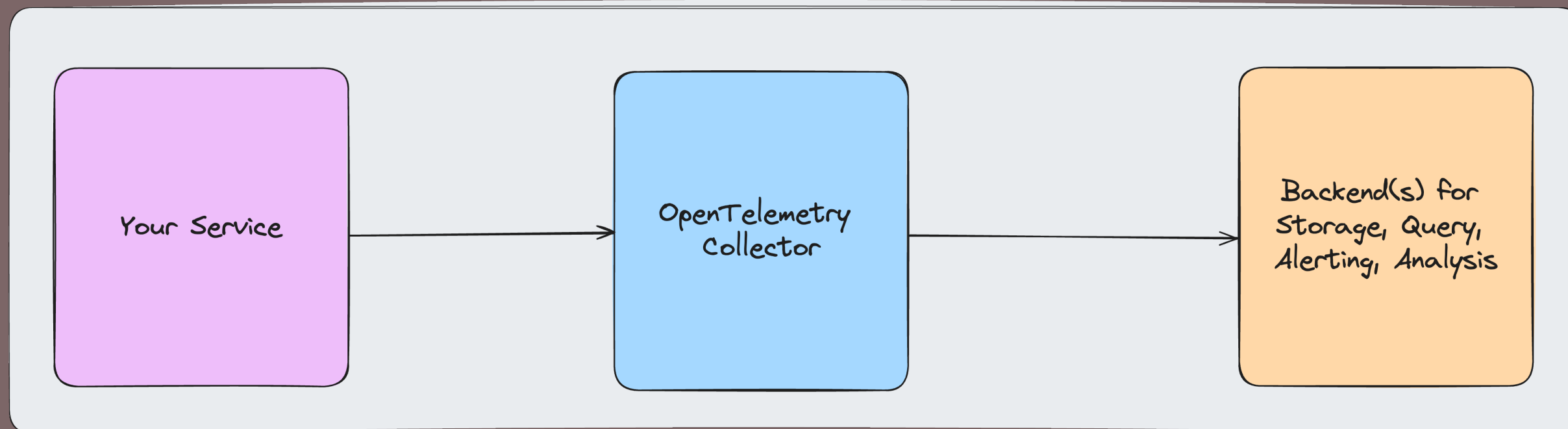
Zero-Code Instrumentation

Go, .NET, Python, PHP,
Java, and JavaScript

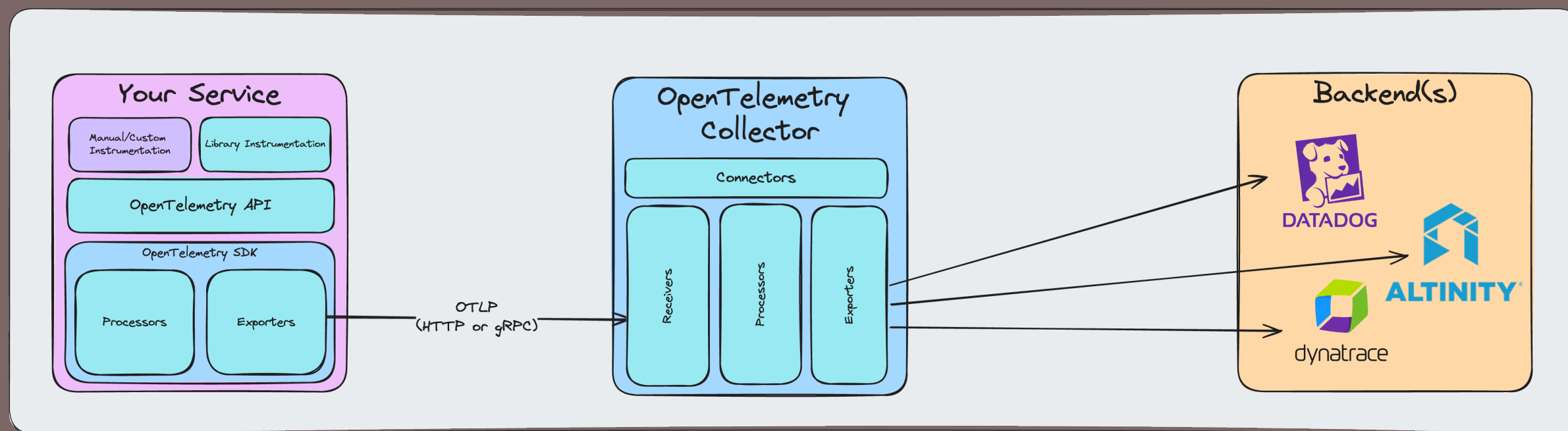
On your node*

OpenTelemetry Collector

OpenTelemetry Collector



OpenTelemetry Architecture



Extensions

(Just a few of my favorites out of 100+)

Receivers	Processors	Exporters	Connectors
OTLP	Batch	OTLP	Span Metrics
Prometheus	OTTL	Prometheus	Service Graph
Kafka	Filter	Jaeger	Exceptions
OpenTracing	Resource Detection	ClickHouse	Routing

New Signals Coming Soon

Profiles, Events, & Errors

Why Open Source Observability?

1. Open-source instrumentation
2. Portable data formats
3. Interoperable tool-chains
4. Growing together as a community

Thank you!

Connect with me:



See you at open spaces!